Name:_____ Period:_____
Earth and Space Science

METAMORPHIC ROCKS
LEARNING GUIDE

Learning Intention
Today we will be learning how heat and pressure change existing rocks without melting them so we can explain why the rocks of the Great Smoky Mountains split into flat sheets.

Success Criteria
I can… ● Explain how heat and pressure change a rock without melting it, and where this happens. ● Tell foliated rocks from nonfoliated rocks and put the rocks that form from shale in order of increasing heat and pressure.

Great Smoky Mountains: Rocks Under Pressure

Great Smoky Mountains National Park, on the border of Tennessee and North Carolina, is the most visited national park in the United States. Its rocks began more than 500 million years ago as layers of mud and sand on an ancient sea floor. Later, when continents collided to build the Appalachian Mountains, those layers were buried, squeezed, and heated. On Day 1, you saw a rock from this park (Site D) that splits into flat sheets. In this guide, the Smokies are our example of how rocks change without melting. As you read, keep this question in mind: *How can a rock turn into a new rock without ever melting?*

Heat and Pressure Without Melting

A **metamorphic rock** forms when an existing rock, called the **parent rock**, is changed by heat and pressure. The rock stays **solid** the whole time. Its atoms rearrange into new or larger crystals, a process called **recrystallization**. If the rock melted, it would become magma and later form an igneous rock instead.

- **Regional metamorphism** happens over large areas when rocks are buried deep and squeezed as plates collide and build mountains. The rocks of the Smokies formed this way.
- **Contact metamorphism** happens in a small area where rock touches hot magma. The rock is "baked" by the heat, but only close to the magma.

Check for Understanding

_____ 1. Which statement best describes how a metamorphic rock forms?
(1) Magma cools and hardens.
(2) Sediments are pressed and cemented together.
(3) An existing rock is changed by heat and pressure without melting.
(4) A rock melts and then recrystallizes.

_____ 2. The rocks of the Great Smoky Mountains formed mainly by
(1) contact metamorphism from a small body of magma
(2) regional metamorphism during a plate collision
(3) lava cooling at Earth's surface
(4) sand and mud settling in water

Foliated and Nonfoliated Rocks

Pressure from one direction squeezes flat minerals, such as mica, until they line up in parallel sheets. This lined-up texture is called **foliation**. Foliated rocks often split into flat pieces. Rocks made mostly of one mineral that does not line up, such as quartz or calcite, are **nonfoliated**: sandstone becomes **quartzite**, and limestone becomes **marble**.

As heat and pressure increase, shale changes into a series of foliated rocks, and its crystals grow larger:

Rock	Heat and pressure	What you observe
Slate	Low	Dull, fine-grained; splits into thin, flat sheets
Phyllite	Low to medium	Splits into sheets with a silky shine from tiny mica crystals
Schist	Medium to high	Sparkly flakes of mica large enough to see
Gneiss	Highest	Minerals separated into light and dark **bands**

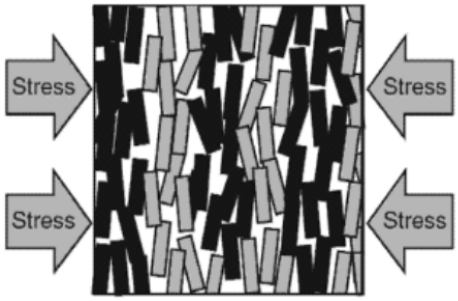
Bands form only at the highest heat and pressure. Most rocks in the Smokies are slate and phyllite, which formed at low heat and pressure, so they split into flat sheets but do not show bands. Bands are also different from the **layers** in sedimentary rock: layers are sediment deposited one on top of another, while bands are minerals that separated during metamorphism.

Check for Understanding: Model It

The diagram represents the minerals in a rock before and after metamorphism.



Mineral Arrangement
Before Metamorphism



Rock C Showing Banding
After Metamorphism

1. **Describe** how the arrangement of the minerals changed after pressure (stress) was applied.

_____

_____

2. Write **F** if the rock is foliated or **N** if it is nonfoliated.

_____ a. Slate

_____ b. Marble

_____ c. Gneiss

_____ d. Quartzite

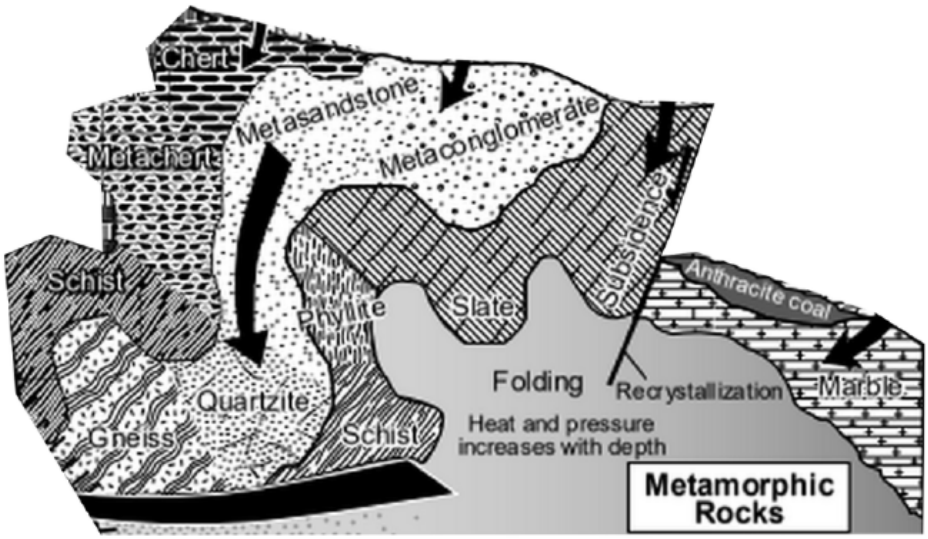
3. Explain why the rocks of the Smokies split into flat sheets but do not show bands.

_____

_____

ESSRT Connection

Use the **metamorphic rocks section** of the **Rock Cycle Infographic** (page 15) in the 2026 Revised Edition Reference Tables for Earth and Space Sciences.



_____ 1. According to the infographic, what happens to heat and pressure as depth increases?
(1) Both decrease.
(2) Both increase.
(3) Heat increases and pressure decreases.
(4) Both stay the same.

_____ 2. Which process labeled on the infographic describes minerals forming new crystals while the rock stays solid?
(1) folding
(2) subsidence
(3) recrystallization
(4) melting

3. The infographic shows sandstone changing into metasandstone. What metamorphic rock forms from it deeper, under more heat and pressure?

_____

4. Gneiss is shown deeper in the infographic than slate. Explain what this tells you about the conditions under which gneiss formed.

_____

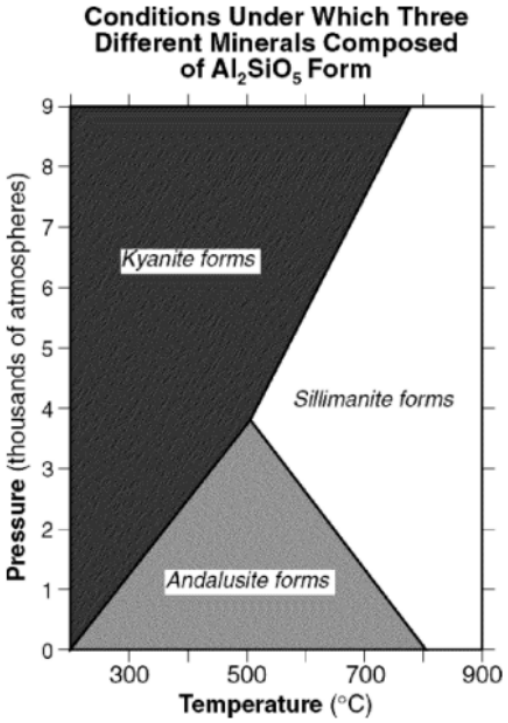
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Cluster Set 1

Base your answers to questions 1 through 4 on the passage and graph below and on your knowledge of Earth and Space Science.

Minerals as Clues to Metamorphic Conditions

Andalusite, kyanite, and sillimanite are three minerals with the same chemical composition, Al_2SiO_5. Like diamond and graphite, their atoms are arranged differently, so they are different minerals. Each one forms only under certain temperature and pressure conditions. When geologists find one of these minerals in a metamorphic rock, they can use a graph like the one below to infer the conditions the rock experienced. Greater pressure means the rock was buried deeper.



_____ 1. Which mineral forms at a temperature of 700°C and a pressure of 2,000 atmospheres?
(1) andalusite
(2) kyanite
(3) sillimanite
(4) all three minerals

2. Which mineral forms at a temperature of 400°C and a pressure of 6,000 atmospheres?

_____

_____ 3. A rock containing andalusite most likely formed under
(1) low pressure, close to Earth's surface
(2) high pressure, deep underground
(3) temperatures above 900°C
(4) conditions where the rock melted

4. A geologist finds kyanite in a metamorphic rock. Using the graph, explain what the kyanite tells the geologist about how deep the rock was buried. Then explain why kyanite and andalusite can be different minerals even though they have the same chemical composition.

_____

_____

_____

Mini Glossary

Bands – light and dark stripes of minerals that separate during high-grade metamorphism (example: gneiss)

Contact metamorphism – metamorphism of a small area of rock heated by nearby magma

Foliation – a texture in which flat minerals are lined up in parallel sheets

Gneiss – a banded metamorphic rock formed at the highest heat and pressure

Marble – a nonfoliated metamorphic rock formed from limestone

Metamorphic rock – a rock formed when an existing rock is changed by heat and pressure without melting

Nonfoliated – having no lined-up minerals (examples: marble, quartzite)

Parent rock – the original rock that is changed into a metamorphic rock

Phyllite – a foliated rock with a silky shine, formed from slate at slightly higher heat and pressure

Quartzite – a nonfoliated metamorphic rock formed from sandstone

Recrystallization – the rearranging of atoms into new or larger crystals while a rock stays solid

Regional metamorphism – metamorphism of large areas during deep burial and plate collisions

Schist – a foliated rock with visible, sparkly flakes of mica

Slate – a dull, fine-grained rock that splits into thin, flat sheets; formed from shale